



Sales Analyst Project Report

Data-Driven Insights & Strategic Recommendations | 2022 - 2023

Analyst

Data Analyst

Tool Stack

Python + PostgreSQL

Period

2022 - 2023

Project Aim & Objectives

Aim

Perform a comprehensive sales analysis on order-level transactional data to uncover actionable revenue insights across product categories, sub-categories, geographic regions, and time dimensions.

Key Objectives

- Identify top 10 revenue-generating products & sub-categories
- Find top 5 best-selling products per geographic region
- Perform month-over-month (MoM) growth comparison: 2022 vs 2023
- Identify highest-sales month per product category
- Determine sub-category with highest profit growth in 2023
- Recommend strategies to improve sales and customer reach

Tools & Technology Stack



Python / Jupyter

Data loading, filtering, cleaning, and preprocessing



PostgreSQL / pgAdmin

Advanced SQL queries, CTEs, and window functions



Pandas

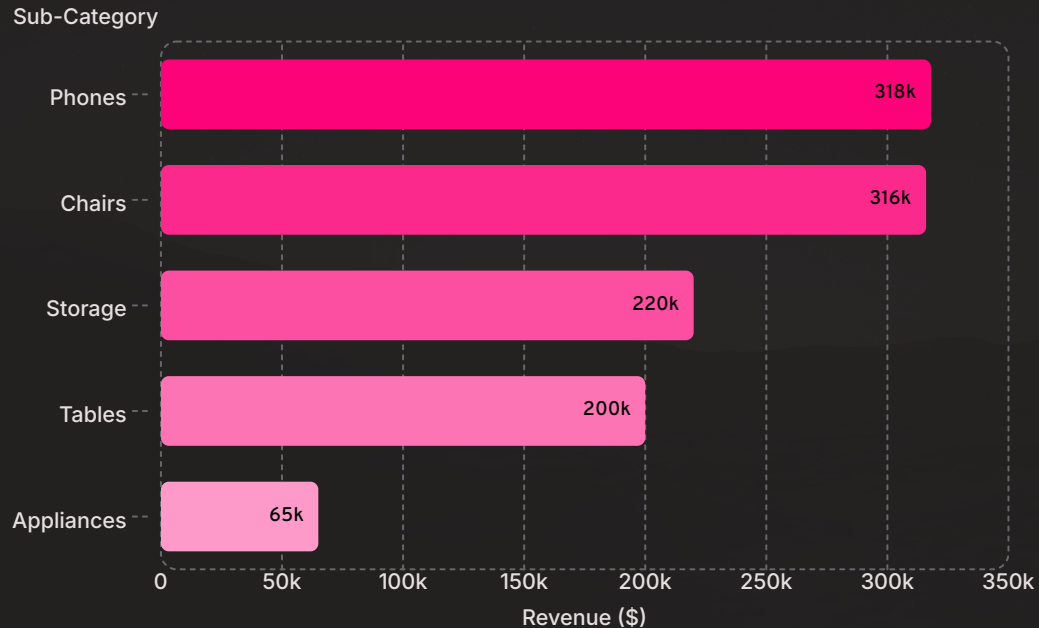
DataFrame manipulation and exploratory data analysis



SQLAlchemy & psycopg2

Python-to-PostgreSQL database connection and adapter

Q1: Top Sub-Categories by Revenue



Key Insight

Phones (\$318K) and Chairs (\$316K) are the top revenue drivers. Technology and Furniture sub-categories dominate the top positions.

Appliances rank 10th, indicating untapped potential. Storage and Tables show strong mid-tier performance worth further investment.

Q2: Top 5 Products per Region

★ Star Product

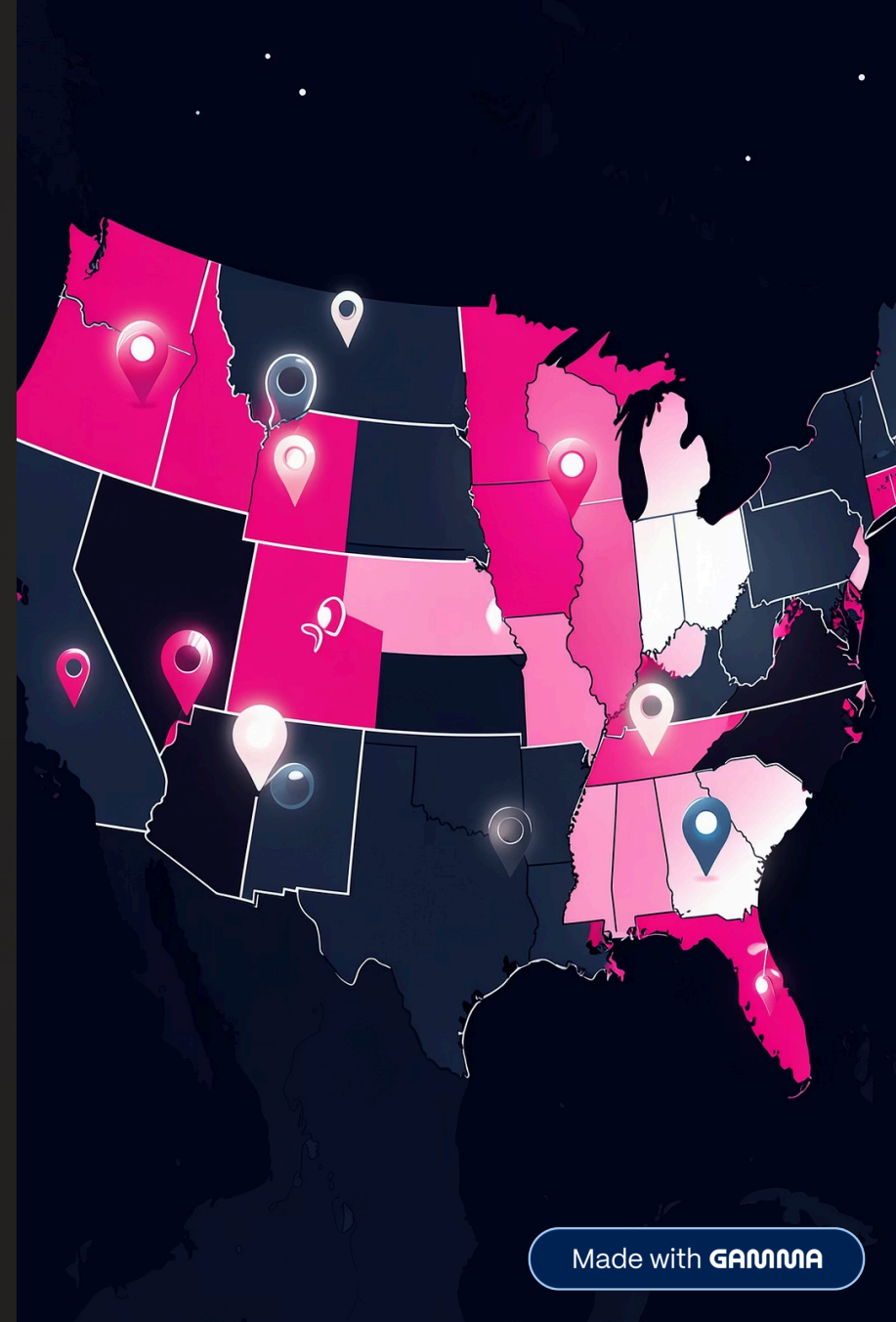
TEC-CO-10004722 ranks #1 across Central, East, and West – 3 out of 4 regions.

East Region

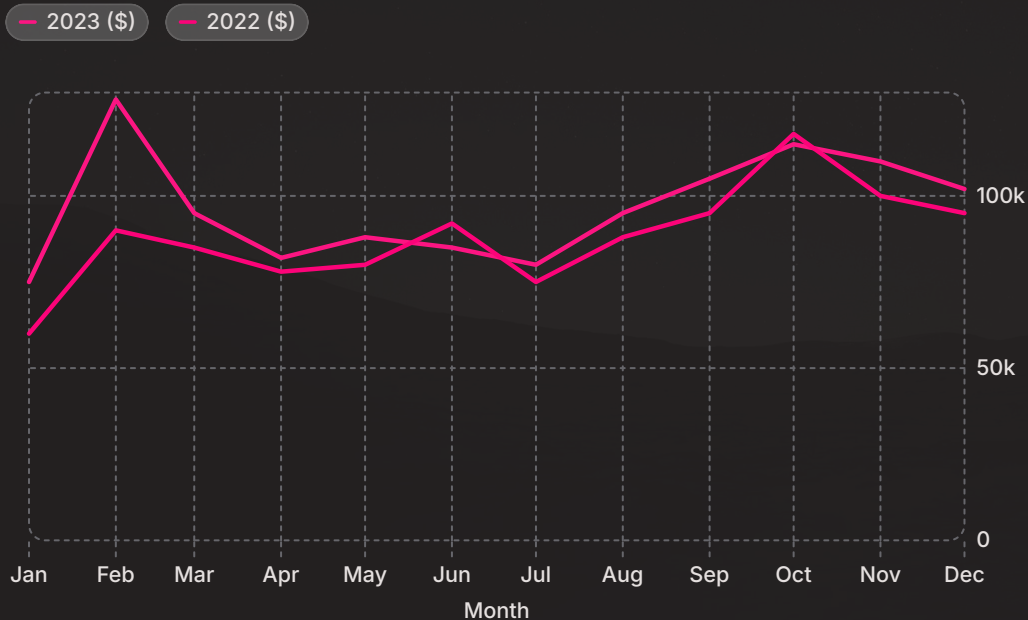
Highest single-product sales at **\$29,099** – the strongest regional performer.

South Opportunity

TEC-CO-10004722 is absent from South's top 5 – a clear expansion opportunity.



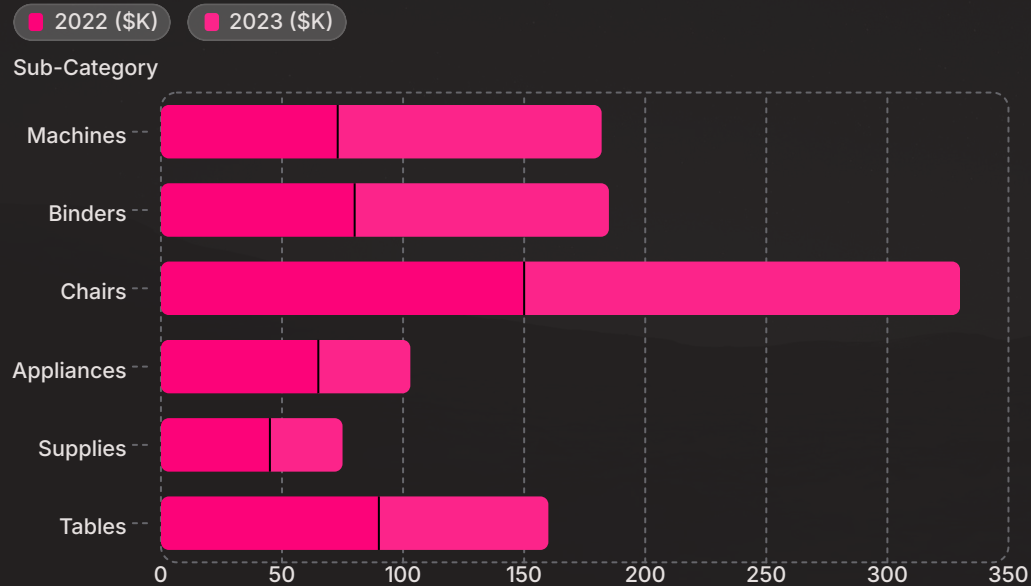
Q3: Month-over-Month Sales – 2022 vs 2023



Key Insights

- **February 2023** surged +42% over 2022 (\$90K → \$128K)
- **October** was the highest 2022 sales month at \$118K
- **June 2023** dipped below 2022 levels – a mid-year slump requiring targeted promotions

Q4: Sub-Category Profit Growth – 2022 vs 2023



Key Insights

- **Machines:** dramatic growth +48% (\$73K → \$109K)
- **Binders & Chairs:** strong positive growth in 2023
- **Appliances:** sharp decline -41% (\$65K → \$38K) – full review needed
- **Supplies & Tables:** contracted – discount strategies may help clear inventory

Strategic Recommendations

1

Product & Category

Double down on Phones & Chairs. Revive Appliances with bundles. Invest in Machines & Binders ahead of Q1/Q4 peaks. Rationalise Supplies & Fasteners with clearance pricing.

2

Regional Strategy

Launch TEC-CO-10004722 in the South. Design region-specific campaigns. Cross-sell tech with accessories. Establish East region warehouses to capitalise on high per-product sales.

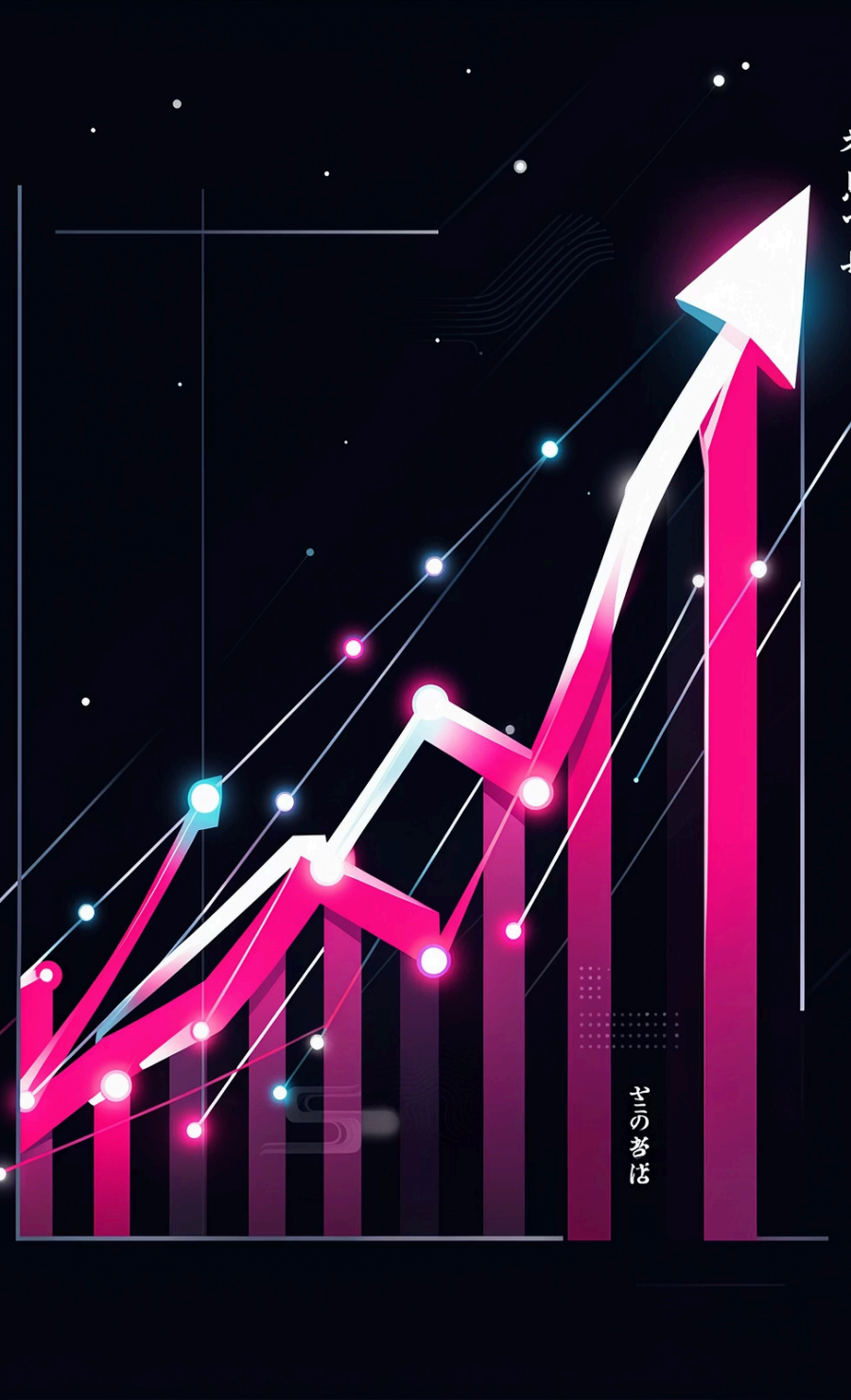
3

Seasonal & Time-Based

Build inventory for October peaks. Address June slump with flash sales & loyalty double-points. Replicate February 2023 campaigns in future Q1s. Run post-holiday January clearance events.

Offer Suggestions to Boost Sales

Offer Type	Target Products	Expected Impact
Bundle Deal	Phones + Accessories / Chairs + Furnishings	Increase avg. order value 20–30%
Clearance Sale (15–25% off)	Appliances, Supplies, Tables	Reduce dead stock, recover cash flow
Buy 2 Get 1 Free	Binders, Fasteners, Labels, Envelopes	Move low-ticket items in high volume
New Customer (10% off)	All categories – first purchase	Grow customer base & market reach
Loyalty Reward Points	Repeat buyers across all sub-categories	Improve retention & lifetime value
Flash Sale (Weekend)	Machines, Bookcases – June/July slump	Counter seasonal slump with urgency
Free Shipping Threshold	Orders above \$500 in Tech/Furniture	Incentivise higher cart value
Seasonal Bundle (Q4 Oct)	Phones, Chairs, Storage – peak season	Capitalise on October sales spike



Conclusion & Key Takeaway

Revenue Leaders

Phones, Chairs, and Machines are clear leaders – prioritise investment here.

Seasonal Patterns

February and October are peak months. June requires intervention with targeted promotions.

Action Areas

Appliances need revival. Regional targeting and bundle offers can unlock significant growth.

Key Takeaway: Data-driven decision making transforms raw transactional data into competitive advantage. Continue quarterly analysis cycles to stay ahead of market trends and customer demands.